Power BI Sales Analytics Dashboard

Project Report

Name:	Sumit Kumar
GitHub:	github.com/1998sumitkumar67-arch
Project Type:	Business Intelligence and Data Analytics
Tool Used:	Power BI Desktop
Dataset:	Sample Superstore Dataset

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Contents

1. Project Overview

This project focuses on analyzing retail sales data using Power BI. The objective is to transform raw transactional data into meaningful business insights through data cleaning, data modeling, DAX calculations, and interactive dashboard development.

The dashboard helps stakeholders monitor sales performance, profitability, customer behavior, and regional trends.

2. Business Objectives

The project aims to answer the following business questions:

- What is the overall sales performance of the company?
- Which regions generate the highest sales?
- Which product categories contribute the most revenue?
- How has sales performance changed over time?
- What is the overall profit margin?
- What insights can support business decision-making?

3. Project Workflow

The project was executed using a structured data analytics workflow to ensure data quality, analytical accuracy, and effective business reporting.

1. Business Understanding
2. Data Collection
3. Data Cleaning and Validation
4. Data Modeling
5. DAX Measure Development
6. Dashboard Design
7. Business Insight Generation
8. Documentation and Reporting
9. GitHub Publication

The workflow followed industry-standard business intelligence practices and helped transform raw data into actionable insights.

4. Dataset Information

The dataset used for this project is the Sample Superstore Dataset. It contains retail transactional data that can be used to analyze sales, profit, customer behavior, product performance, and regional trends.

Table 4.1: Dataset Summary

Attribute	**Details**
Dataset Name	Sample Superstore Dataset
Number of Rows	9,994
Number of Columns	21

4.1 Key Fields

- Order Date
- Ship Date
- Product Name
- Category
- Sub-Category
- Region
- Sales
- Profit
- Quantity
- Customer Information

5. Data Cleaning and Validation

The following data preparation steps were performed using Power Query:

1. Imported the dataset into Power BI.
2. Verified and corrected data types.
3. Converted date fields to proper date format.
4. Performed data profiling using:
    - Column Quality
    - Column Distribution
    - Column Profile
5. Checked for null values.
6. Checked for data errors.
7. Checked for duplicate records.
8. Validated categorical fields.
9. Confirmed dataset quality before modeling.

5.1 Data Cleaning Result

- No missing values found.
- No duplicate rows found.
- No data quality issues detected.

6. Data Modeling

A dedicated Date Table was created to support time intelligence and improve model performance.

6.1 Relationship Created

`DateTable[Date]` → `Orders[Order Date]`

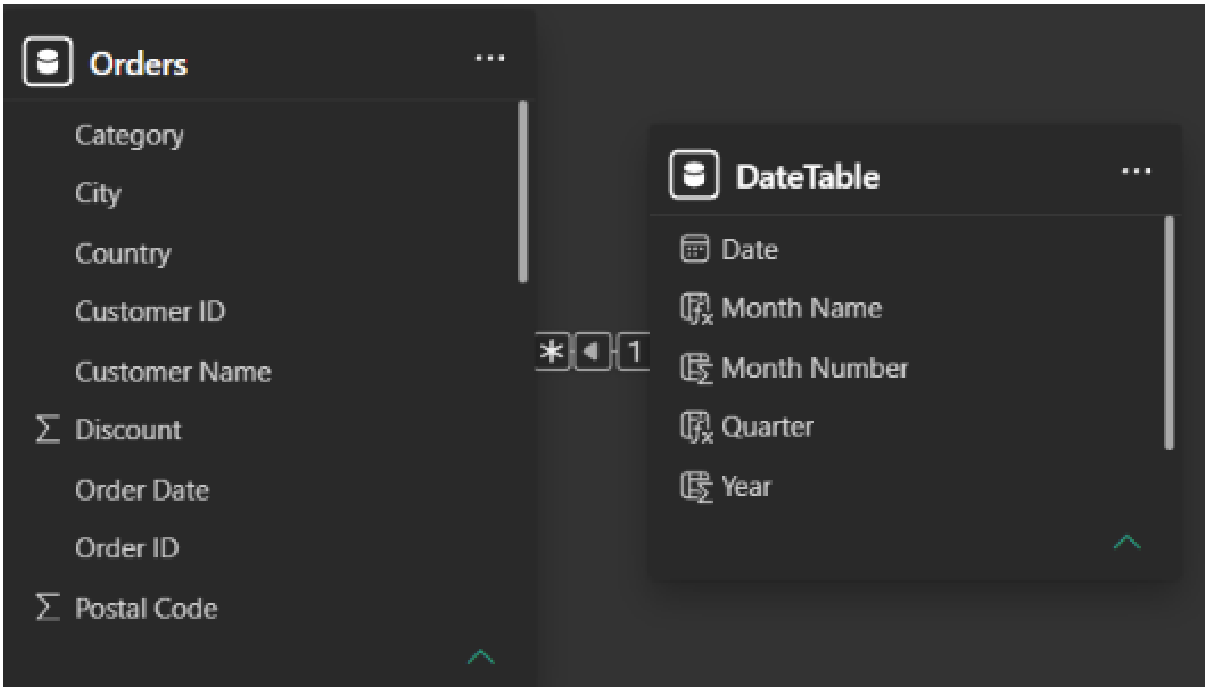

Figure 6.1: Data Model

6.2 Model Type

The data model follows a **Star Schema** design. This structure improves reporting performance and makes the model easier to understand and maintain.

6.3 Date Attributes Created

- Year
- Month Number
- Month Name
- Quarter
- Year Month

7. DAX Measures

The following DAX measures were created to calculate key business metrics:

Table 7.1: DAX Measures

Measure Name	Purpose
Total Sales	Calculates overall sales revenue
Total Profit	Calculates total profit generated
Total Orders	Counts the total number of orders
Total Quantity	Calculates total quantity sold
Profit Margin %	Measures profit as a percentage of sales
Average Order Value	Calculates average revenue per order

7.1 Sample DAX Implementation

- `Total Sales = SUM(Orders[Sales])`
- `Total Profit = SUM(Orders[Profit])`
- `Profit Margin % = DIVIDE([Total Profit],[Total Sales])`

7.2 KPI Results

The following table summarizes the key performance indicators derived from the analysis:

Table 7.2: KPI Results

KPI	Value
Total Sales	$2.30M
Total Profit	$286K
Total Orders	5009
Profit Margin	12.47%
Avg Order Value	$458.61

8. Dashboard Development

The dashboard was developed using Power BI Desktop and designed to provide decision-makers with a clear and interactive view of sales performance.

8.1 Current Development Status

Completed Dashboard Pages:

- Executive Overview Dashboard
- Product Performance Analysis Dashboard

Planned Dashboard Pages:

- Customer Analysis Dashboard

8.2 Dashboard Preview

Figure 8.1 shows the Executive Overview dashboard developed in Power BI.

8.3 Product Performance Analysis Page

The Product Performance Analysis page provides detailed insights into product-level performance. It enables users to identify top-selling products, high-profit products, and profitable product categories through interactive visualizations and filters.

8.4 Executive Overview Page

The Executive Overview page was designed to provide a high-level summary of business performance. It includes key performance indicators and visualizations that allow users to quickly understand sales trends, profitability, and regional performance.

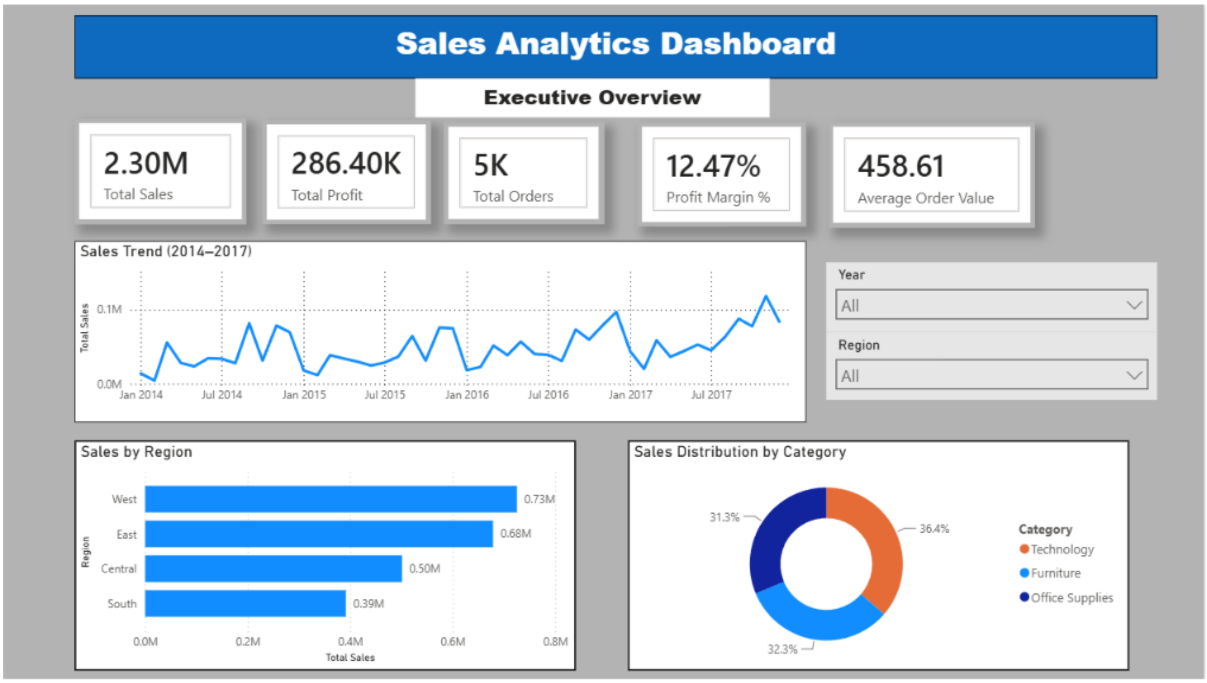

Figure 8.1: Executive Overview Dashboard

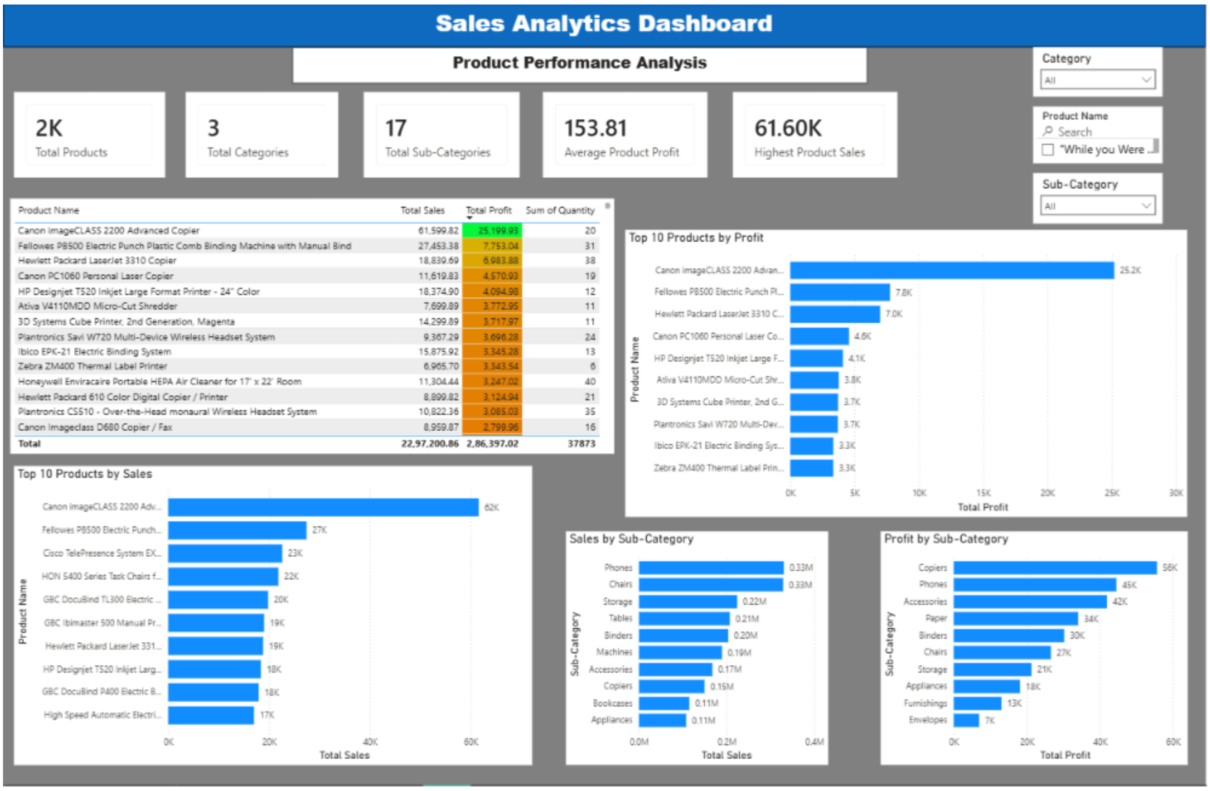

Figure 8.2: Product Performance Analysis Dashboard

8.5 Key Performance Indicators

- Total Sales

- Total Profit
- Total Orders
- Profit Margin %
- Average Order Value

8.6 Executive Overview Visualizations

- Monthly Sales Trend
- Sales by Region
- Sales by Category
- Year Filter
- Region Filter

8.7 Product Performance Visualizations

- Top 10 Products by Sales
- Top 10 Products by Profit
- Sales by Sub-Category
- Profit by Sub-Category
- Product Performance Details Table
- Category Filter
- Sub-Category Filter
- Product Search Filter

9. Key Insights

The analysis of the Sample Superstore dataset revealed several important business insights:

1. The company generated total sales of approximately \$2.30 Million, indicating a strong overall revenue performance.

2. Total profit exceeded \$286 Thousand, resulting in a healthy profit margin of 12.47%.

3. The West region was the highest-performing region, contributing approximately 32% of total sales and demonstrating strong market demand.

4. The Technology category emerged as the leading revenue contributor, accounting for more than one-third of overall sales.

5. The Furniture and Office Supplies categories also contributed significantly to revenue, highlighting a diversified product portfolio.

6. Sales exhibited an overall upward trend between 2014 and 2017, indicating sustained business growth over time.

7. The average order value of \$458.61 suggests that customers typically place moderately high-value orders.

8. Regional and category-level analysis revealed opportunities to optimize sales strategies and improve profitability across different business segments.

10. Business Recommendations

Based on the analysis, the following recommendations can be considered:

- Focus marketing efforts on the West region due to strong sales performance.
- Expand high-performing technology products.
- Monitor low-performing categories and optimize inventory.
- Continue tracking profitability trends using interactive dashboards.
- Invest in high-margin product categories.
- Use customer segmentation to improve targeted marketing campaigns.

11. Tools Used

- Power BI Desktop
- Power Query
- DAX
- Data Modeling
- GitHub

12. Skills Demonstrated

This project demonstrates practical data analytics and business intelligence skills.

- Data Cleaning
- Data Validation
- Power Query
- Data Modeling
- Star Schema Design
- DAX Calculations
- Dashboard Design
- Data Visualization
- Business Analysis
- Report Writing
- GitHub Project Management
- Product Performance Analysis
- Conditional Formatting
- Interactive Filtering
- Dashboard Navigation

13. Project Outcomes

The project successfully achieved the following outcomes:

- Developed an interactive Power BI dashboard.
- Implemented a star schema data model.
- Created reusable DAX measures.
- Improved analytical reporting capabilities.
- Identified key business trends and performance indicators.
- Generated actionable business insights.
- Strengthened practical Power BI and business intelligence skills.

14. Challenges Faced and Solutions

During the development of this project, several technical and analytical challenges were encountered. These challenges helped improve problem-solving skills and provided valuable learning experiences.

14.1 Date Conversion Errors

Challenge: During data import, date fields generated parsing and formatting errors due to inconsistent date formats.

Solution: The issue was resolved by validating and correcting data types in Power Query and ensuring all date fields were converted to the appropriate Date format before data modeling.

14.2 Data Modeling Challenges

Challenge: Building an efficient data model for time-based analysis required proper relationship management and support for DAX time intelligence functions.

Solution: A dedicated Date Table was created and connected to the Orders table using a one-to-many relationship. A Star Schema design was implemented to improve model performance and maintainability.

14.3 Dashboard Design Challenges

Challenge: Designing an interactive dashboard while maintaining readability and visual consistency required multiple layout adjustments.

Solution: Dashboard visuals were organized into logical sections, interactive slicers were added, and visual formatting was optimized to improve user experience and analytical effectiveness.

14.4 Key Learnings

- Importance of data quality before analysis.
- Practical implementation of Star Schema modeling.
- Creation and optimization of DAX measures.
- Dashboard design best practices in Power BI.
- GitHub project documentation and portfolio management.

15. Future Enhancements

The project can be further improved by adding the following features:

- Customer Analysis Dashboard
- Profitability Analysis Dashboard
- Regional Performance Dashboard
- Advanced DAX Measures
- Drill-through Reports

16. References

- Microsoft Power BI Documentation
- Microsoft Learn Power BI Learning Path
- Sample Superstore Dataset
- Power BI Community Documentation

17. Project Repository

The complete project including dashboard files, documentation, screenshots, and future updates is maintained on GitHub. https://github.com/1998sumitkumar67-arch/PowerBI-sales-analytics-dashboard

18. Conclusion

This project demonstrates the complete Power BI analytics workflow, including data preparation, data modeling, DAX calculations, dashboard development, and business insight generation. The dashboard provides decision-makers with an interactive view of sales performance and business trends.